

3. A children's swimming pool is being filled with water. The amount of water in the pool at any time can be found using the equation $A = 20t + 5$, where A is the amount of water in litres and t is the time in minutes.
- Explain why this equation can be represented by a straight line.
 - State the value of the y -intercept and what it represents.
 - Sketch the graph of $A = 20t + 5$ on a set of axes.
 - The pool holds 500 litres. By solving an equation, determine how long it will take to fill the swimming pool. Write your answer correct to the nearest minute.
4. **WEB** A yoga ball is being pumped full of air at a rate of $40 \text{ cm}^3/\text{second}$. Initially there is 100 cm^3 of air in the ball.
- Construct an equation that represents the amount of air, A , in the ball after t seconds.
 - Interpret what the value of the y -intercept in the equation means.
 - How much air, in cm^3 , is in the ball after 2 minutes?
 - When fully inflated the ball holds $100\,000 \text{ cm}^3$ of air. Determine how long, in minutes, it takes to fully inflate the ball. Write your answer to the nearest minute.

